

ThmSmith Stage -1 proposal:
q1_spectral_phase_transition / p1_markov
*Markov switch scarcity turns Sun–Abraham contamination into
event-window rank collapse*

ThmSmith Stage -1 proposer

May 13, 2026

Locked parameters.

qid = q1_spectral_phase_transition
specialization = p1_markov
p = 1
assignment = stationary two-state first-order Markov treatment path with
transition probabilities $a_T = \Pr(D_t=1 \mid D_{t-1}=0)$ and
 $b_T = \Pr(D_t=0 \mid D_{t-1}=1)$. The spectral gap is
 $\gamma_T = a_T + b_T$, the second eigenvalue is $\rho_T = 1 - \gamma_T$,
and the stationary switch-on flow is $\phi_T = a_T b_T / (a_T + b_T)$.
 $D_{i0}=0$ is a synthetic lag used only to classify the first history.
target = finite-window switch-on event profile θ_k , k in K , where an
event origin s satisfies $H_{is}=(0,1)$, θ_0 is the switch-on ATT
 $m_s(01) - m_s(00)$, and θ_k for $k>0$ is the survivor-window contrast
 $m_{s+k}(11) - m_{s+k}(00)$ conditional on no off-switch between s and $s+k$.
regression = TWFE/FWL event-window regression with unit and calendar-time
fixed effects, switch-on-origin relative-time columns E_k , k in K
(one omitted pre-period), and the Q1 clean history nuisance columns
 $1\{H=(1,0)\}$ and $1\{H=(1,1)\}$. Regimes with singular values below the
deterministic threshold κ_T are interpreted through the regular
row-space and bounded-response affine-fiber convention in Assumption 5.

Abstract

Sun and Abraham show that conventional event-study coefficients
can be contaminated by other relative periods when treatment effects
are heterogeneous. This proposal asks whether Markov persistence cre-
ates a sharper phenomenon: relative-period switch-on contrasts may

cease to be regularly estimable because the isolated event-window rows disappear from the FWL row space at the relevant rate. The proposed kernel is a threshold theorem in the effective clean-window count $\Lambda_T(K) = T\phi_T(1 - a_T)^{K-}(1 - b_T)^{K+}$, which ties event-window support to the Markov spectral parameters. Above the threshold, the event-window ATT profile is regularly point identified; on the boundary, only the projection onto the retained row space is regular and the remaining coordinates are bounded; below it, the target is non-regularly identified and unbounded response surfaces are not point identified. The flagship statement is a Conjecture, with a routine linear-algebra theorem isolating exactly what Stage 0 would need to prove once the Markov singular-value asymptotics are established.

1 Introduction

Why the problem is important. Event-study DiD is often reported as a dynamic profile, so losing one relative-period coordinate is qualitatively worse than merely changing a weight inside an average. The modern staggered-adoption literature explains contamination under heterogeneous effects, but it usually assumes the relative-time rows exist and then asks whether the wrong rows contaminate the coefficient. In a Markov treatment panel with rare switching or strong persistence, the empirical problem can be more basic: the switch-on windows that define relative time may be too scarce for the FWL residualized design to retain full row rank at the rate needed for regular identification.

The main finding (proposed). The headline conjecture is Conjecture 1: for any fixed finite event window $K = \{-K_-, \dots, K_+\}$ with one omitted pre-period, the threshold-retained row space of the TWFE/FWL event-window design is governed by the effective clean-window count

$$\Lambda_T(K) = T \frac{a_T b_T}{a_T + b_T} (1 - a_T)^{K-} (1 - b_T)^{K+}.$$

If $\Lambda_T(K)/\kappa_T \rightarrow \infty$, all non-omitted event-window switch-on contrasts are regularly visible; if $\Lambda_T(K)/\kappa_T \rightarrow \lambda \in (0, \infty)$, the visible row space is lower-dimensional and the invisible coordinates enter a bounded affine fiber; if $\Lambda_T(K)/\kappa_T \rightarrow 0$, the event-window profile is not regularly identified without additional shape restrictions.

What is new. The closest published paper is [Sun and Abraham \(2021\)](#), which establishes contamination of event-study leads and lags under hetero-

geneous effects. The new kernel is not another contamination decomposition and not an extension from absorbing to Markov assignment for its own sake. It claims that Markov persistence turns contamination into a rank-collapse problem: there are regimes in which a relative-period column is present in the regression formula but its target row falls outside the threshold-retained FWL row space.

How it positions in the literature. The proposal is a bridge between event-study contamination, switchback/panel-experiment serial dependence, and partial identification. It uses the Q1 minimal-basis language already present in the repository, but its novelty axis is the Sun–Abraham event-window row space rather than the sign of Markov FWL weights. It is closest to [Bojinov, Rambachan, and Shephard \(2021\)](#) and [Hu and Wager \(2022\)](#) in treating serial dependence as a first-order design feature, while [Borusyak, Jaravel, and Spiess \(2024\)](#) and [de Chaisemartin and D’Haultfoeuille \(2024\)](#) provide the nearest DiD estimand comparisons.

2 Related work

[Sun and Abraham \(2021\)](#) show that event-study lead and lag coefficients in staggered designs can be contaminated by treatment effects from other relative periods; this proposal asks when Markov switch scarcity makes some relative-period rows non-regularly estimable at all. [de Chaisemartin and D’Haultfoeuille \(2020\)](#) show that TWFE coefficients with heterogeneous effects are weighted averages with possibly negative weights; the present proposal keeps the weighted FWL representation but studies thresholded row rank rather than signs. [Goodman-Bacon \(2021\)](#) decomposes staggered timing TWFE into two-by-two comparisons; the proposed Markov event-window threshold is a case where the comparison labels can remain formally present while the FWL row span loses dimension. [Callaway and Sant’Anna \(2021\)](#) identify group-time ATT components and aggregate them under staggered adoption assumptions; the proposed result is an analogous visibility diagnostic for Markov history-time ATT components before aggregation. [Borusyak, Jaravel, and Spiess \(2024\)](#) derive an efficient imputation estimator in staggered event studies when untreated outcomes for target cells are spanned by observed untreated information; the row-span language in Theorem 1 is the FWL counterpart of that support condition. [de Chaisemartin and D’Haultfoeuille \(2024\)](#) study DiD estimators for nonabsorbing and lagged treatments; the proposed theorem is narrower algebraically but

sharper about the Markov spectral support boundary of a TWFE/FWL event-window basis. [Bojinov, Rambachan, and Shephard \(2021\)](#) analyze panel experiments with dynamic causal effects and show that serially correlated assignment undermines conventional fixed-effect causal interpretations; this proposal turns that warning into a concrete row-rank threshold for event-window ATT profiles. [Bojinov, Simchi-Levi, and Zhao \(2023\)](#) derive design and inference results for switchback experiments with carryover; the present proposal differs by studying when a Markov switchback-style path law makes TWFE event-window rows visible or invisible. [Hu and Wager \(2022\)](#) prove that geometrically mixing Markovian systems can change switchback estimation rates; this is the closest rate phenomenon, but it does not state a Sun–Abraham-style relative-period row-space phase transition. [Han \(2021\)](#) studies identification in nonparametric dynamic treatment models; the proposal is more specialized because the only dynamic assignment law is a two-state Markov chain and the object is the FWL event-window row space. [Athey and Imbens \(2022\)](#) give a design-based analysis of staggered adoption under randomized adoption timing; the proposed Markov law replaces randomized absorbing adoption dates with recurrent switch-on origins whose frequency is controlled by ϕ_T and ρ_T . [Rambachan and Roth \(2023\)](#) develop partial-identification sensitivity analysis for deviations from parallel trends; here partial identification arises even under the exact untreated TWFE baseline because the event-window row is not retained at the regularity threshold. [Manski \(2003\)](#) and [Tamer \(2010\)](#) supply the generic identified-set logic used for bounded affine fibers; the new content is the Markov/FWL condition determining when that logic becomes necessary. In the repository, `AutoID/Panel/Theorems/Q1_MinimalBasis.lean` proves the clean Q1 history-basis substrate; `AutoID/Panel/Theorems/Q1_StaggeredMinimalBasis.lean` develops a staggered analogue; and `AutoID/Panel/Theorems/Q1_SpectralThreshold/*.lean` proves a Markov sign-threshold refutation with explicit $T = 3$ FWL weights and long- T limits. The relevant API entries are `doc/API.md` section 14 for `Panel/`, especially the FWL residualization, saturated-history, finite weighted algebra, and partial-identification infrastructure.

3 Formal setup

Let $i = 1, \dots, N$, $t = 1, \dots, T$, and $D_{it} \in \{0, 1\}$. The first-order history is $H_{it} = (D_{i,t-1}, D_{it}) \in \mathcal{H}_1 = \{00, 01, 10, 11\}$, with the synthetic boundary $D_{i0} = 0$. Potential outcomes have memory $p = 1$: $Y_{it} = Y_{it}(H_{it})$. Write $m_t(h) = \mathbb{E}[Y_{it}(h)]$, $R_{it} = \mathbf{1}\{H_{it} = 01\}$, $C_{10,it} = \mathbf{1}\{H_{it} = 10\}$, and $C_{11,it} =$

$\mathbf{1}\{H_{it} = 11\}$. The untreated baseline is $\mathbb{E}[Y_{it}(00) \mid D_i] = \alpha_i + \lambda_t$.

A switch-on origin is a cell s with $H_{is} = 01$. For an event window $K = \{-K_-, \dots, K_+\}$, define $W_{is}^K = 1$ if $H_{is} = 01$, $D_{i,s-r} = 0$ for $1 \leq r \leq K_-$, and $D_{i,s+r} = 1$ for $1 \leq r \leq K_+$; otherwise $W_{is}^K = 0$. Thus W^K records an isolated clean switch-on window: no on-switch occurs in the K_- pre-window and no off-switch occurs in the K_+ post-window. The event-window ATT coordinates are

$$\theta_0 = m_s(01) - m_s(00), \quad \theta_k = m_{s+k}(11) - m_{s+k}(00) \quad (1 \leq k \leq K_+),$$

with calendar averaging over origins satisfying $W^K = 1$. Pre-period coefficients are diagnostic rows rather than causal effect targets; one pre-period is omitted from the regression.

The Markov assignment has transition matrix

$$M_T = \begin{pmatrix} 1 - a_T & a_T \\ b_T & 1 - b_T \end{pmatrix}, \quad \gamma_T = a_T + b_T, \quad \rho_T = 1 - \gamma_T, \quad \phi_T = \frac{a_T b_T}{a_T + b_T}.$$

Under stationarity, the clean-window probability is

$$p_T(K) = \phi_T(1 - a_T)^{K_-}(1 - b_T)^{K_+}, \quad \Lambda_T(K) = T p_T(K).$$

The TWFE/FWL event-window regression contains unit fixed effects, calendar-time fixed effects, event-window columns E_k generated by switch-on origins $W^K = 1$, $k \in K$ except the omitted pre-period, and the Q1 clean nuisance columns C_{10} and C_{11} . Let $X_T(K)$ be this residualized regressor matrix after projecting on fixed effects. Let $\Phi_T(K)$ be the residualized local response-surface cell basis for the histories and event-window offsets appearing in K . Define the FWL moment map $A_T(K) = \mathbb{E}[X_T(K)^\top \Phi_T(K)]$, the Gram matrix $G_T(K) = \mathbb{E}[X_T(K)^\top X_T(K)]$, and the response vector m_K collecting the local $m_{s+k}(h)$ coordinates. For a target row ℓ_K , the event-window target is $\theta_{\ell,K} = \ell_K^\top m_K$.

For a deterministic regularity threshold $\kappa_T > 0$, define the retained singular row space

$$\mathcal{V}_T^{\text{reg}}(K) = \text{span}\{v_j^\top A_T(K) : \sigma_j(G_T(K)) \geq \kappa_T\},$$

where v_j is a left singular direction of $X_T(K)$ and $\sigma_j(G_T(K))$ is the associated Gram singular value. Let $\Pi_T^{\text{reg}}(K)$ be the Euclidean projector onto $\mathcal{V}_T^{\text{reg}}(K)$. The bounded-response affine fiber at observed FWL moment $z = A_T(K)m_K$ is

$$\mathcal{F}_T(z; L, U) = \{m' : A_T(K)m' = z, \ L \leq m'_j \leq U \text{ for every coordinate } j\}.$$

Locked parameters for Stage 0:

`qid = q1_spectral_phase_transition`
`specialization = p1_markov`
`p = 1`
`assignment = stationary two-state first-order Markov treatment path with`
`transition probabilities  $a_T = \Pr(D_t=1 \mid D_{t-1}=0)$  and`
 `$b_T = \Pr(D_t=0 \mid D_{t-1}=1)$ . The spectral gap is`
 `$\gamma_T = a_T + b_T$ , the second eigenvalue is  $\rho_T = 1 - \gamma_T$ ,`
`and the stationary switch-on flow is  $\phi_T = a_T b_T / (a_T + b_T)$ .`
 `$D_{i0}=0$  is a synthetic lag used only to classify the first history.`
`target = finite-window switch-on event profile  $\theta_k$ ,  $k$  in  $K$ , where an`
`event origin  $s$  satisfies  $H_{is}=(0,1)$ ,  $\theta_0$  is the switch-on ATT`
 `$m_s(01) - m_s(00)$ , and  $\theta_k$  for  $k>0$  is the survivor-window contrast`
 `$m_{s+k}(11) - m_{s+k}(00)$  conditional on no off-switch between  $s$  and  $s+k$ .`
`regression = TWFE/FWL event-window regression with unit and calendar-time`
`fixed effects, switch-on-origin relative-time columns  $E_k$ ,  $k$  in  $K$`
`(one omitted pre-period), and the Q1 clean history nuisance columns`
 `$1\{H=(1,0)\}$  and  $1\{H=(1,1)\}$ . Regimes with singular values below the`
`deterministic threshold  $\kappa_T$  are interpreted through the regular`
`row-space and bounded-response affine-fiber convention in Assumption 5.`

4 Assumptions

Assumption 1 (Finite memory and consistency). For every unit and period, the observed outcome is $Y_{it} = Y_{it}(H_{it})$, where $H_{it} = (D_{i,t-1}, D_{it})$.

Assumption 2 (Untreated TWFE baseline). For the untreated history, $\mathbb{E}[Y_{it}(00) \mid D_i] = \alpha_i + \lambda_t$ for some unit and calendar-time functions α_i and λ_t .

Assumption 3 (Stationary Markov assignment). For each T , $D_{i1}, \dots, D_{iT}$ are drawn from the stationary two-state Markov law with transition probabilities $0 < a_T < 1$ and $0 < b_T < 1$, independently across units.

Assumption 4 (Clean switch-on event window). The event-window columns E_k are generated only from origins satisfying $W_{is}^K = 1$, with one pre-period omitted and with K_-, K_+ fixed or growing deterministically with T .

Assumption 5 (Regular row-space convention). A deterministic threshold $\kappa_T > 0$ is fixed before observing outcomes. Linear functionals in $\mathcal{V}_T^{\text{reg}}(K)$ are called regularly identified; coordinates outside that row space are treated through the affine fiber $\mathcal{F}_T(z; L, U)$ under finite constants $L < U$.

Assumption 6 (Bounded response surface). For partial-identification statements, every local response coordinate in m_K lies in the known interval $[L, U]$, where $L < U$.

Assumption 7 (Nondegenerate target row). The target row ℓ_K is nonzero and has support only on the switch-on and survivor-window contrasts defined in the formal setup.

5 Main results

Theorem 1 (Theorem 1: regular row-space trichotomy). *Assume Assumptions 1, 2, 5, 6, and 7. Fix T , K , an observed FWL moment $z = A_T(K)m_K$, and target row ℓ_K . If $\ell_K \in \mathcal{V}_T^{\text{reg}}(K)$, then $\theta_{\ell,K} = \ell_K^\top m_K$ is regularly point identified from z . If $0 \neq \Pi_T^{\text{reg}}(K)\ell_K \neq \ell_K$, then the projected target $(\Pi_T^{\text{reg}}(K)\ell_K)^\top m_K$ is regularly point identified and the full target has the sharp bounded identified interval*

$$\left[\inf_{m' \in \mathcal{F}_T(z; L, U)} \ell_K^\top m', \sup_{m' \in \mathcal{F}_T(z; L, U)} \ell_K^\top m' \right].$$

If $\Pi_T^{\text{reg}}(K)\ell_K = 0$, then no nonzero component of $\theta_{\ell,K}$ is regularly identified; without Assumption 6, the target is not point identified whenever the affine fiber contains a direction v with $A_T(K)v = 0$ and $\ell_K^\top v \neq 0$.

Why this is new and worth studying. The algebra is a finite-dimensional row-space fact, so it is a Theorem rather than the novelty claim. Its gap to [Sun and Abraham \(2021\)](#) and [Borusyak, Jaravel, and Spiess \(2024\)](#) is that it isolates the exact regular row-space object needed for a Markov support threshold; resolving the conjecture below would turn event-study support diagnostics into explicit point/partial/nonregular identification statements.

Conjecture 1 (Conjecture 1: Markov switch-scarcity event-window phase transition (NOT YET PROVED)). *Under Assumptions 1–7, suppose $K = K_T$ is deterministic and $p_T(K) > 0$. There are constants $0 < c_K < C_K < \infty$, depending only on the event-window template and the omitted pre-period convention, such that the nonzero singular values of the event-window FWL Gram block generated by E_k, C_{10}, C_{11} satisfy*

$$c_K \Lambda_T(K) \leq \sigma_j(G_T(K)) \leq C_K \Lambda_T(K)$$

for every event-window direction whose clean switch-on row is represented, while every direction orthogonal to the clean-window row space has singular

value $o(\Lambda_T(K))$. Consequently, if $\Lambda_T(K)/\kappa_T \rightarrow \infty$, then $\ell_K \in \mathcal{V}_T^{\text{reg}}(K)$ for every finite-window switch-on ATT loading ℓ_K ; if $\Lambda_T(K)/\kappa_T \rightarrow \lambda \in (0, \infty)$, only the projection onto the limiting clean-window row space is regular; and if $\Lambda_T(K)/\kappa_T \rightarrow 0$, no nonzero switch-on event-window loading is regularly identified.

Why this is new and worth studying. The closest paper, [Sun and Abraham \(2021\)](#), says relative-period coefficients can load on the wrong treatment-effect rows; it does not claim that Markov persistence creates a singular-value boundary where the rows disappear from the regular design. [Hu and Wager \(2022\)](#) connect Markov mixing to rates in switchback experiments, but not to a TWFE/FWL event-study row space. If true, this conjecture gives practitioners a reportable diagnostic: before interpreting a dynamic ATT profile, compute the effective clean-window count and declare which coordinates are point identified, bounded, or non-regular.

Conjecture 2 (Conjecture 2: minimal-basis structural change at the boundary (NOT YET PROVED)). *Under the same assumptions as Conjecture 1, the Q1 nuisance basis $\{C_{10}, C_{11}\}$ is minimal for regular point identification of all finite-window switch-on event contrasts when $\Lambda_T(K)/\kappa_T \rightarrow \infty$. On the boundary $\Lambda_T(K)/\kappa_T \rightarrow \lambda \in (0, \infty)$, the minimal regular basis is the quotient of $\{C_{10}, C_{11}\}$ by the null nuisance contrast generated by the clean-window row-space kernel. Below the boundary, no finite augmentation by relative-time columns alone restores regular identification of the full event-window profile unless it adds an external shape restriction on m_K .*

Why this is new and worth studying. The in-repository Q1 minimal-basis theorems treat the clean history basis as a static algebraic object. This conjecture proposes that the minimal basis itself changes across a Markov support threshold, a phenomenon not covered by [de Chaisemartin and D’Haultfoeuille \(2024\)](#) or [Bojinov, Rambachan, and Shephard \(2021\)](#). Resolving it would connect formula design directly to the observed assignment spectrum rather than treating regression specification as fixed.

6 Sanity-check corollaries

Corollary 1 (IID fixed-window reduction). *If $a_T = a$, $b_T = b$, $0 < a, b < 1$, and K is fixed, then $\Lambda_T(K) \asymp T$. For any threshold $\kappa_T = o(T)$, Conjecture 1 reduces to regular point identification of the finite switch-on event-window profile.*

This reduces cleanly under Assumptions 3 and 4: $p_T(K)$ is a fixed positive constant, so the local Gram block grows linearly in T , matching the existing Q1 clean-basis intuition that all four histories have stable support.

Corollary 2 (Rare-switch failure). *If $a_T = \alpha/T^2$, $b_T = b \in (0, 1)$, and K is fixed, then $\Lambda_T(K) \rightarrow 0$. Conjecture 1 therefore predicts that no nonzero switch-on event-window loading is regularly identified at any fixed positive threshold.*

The reduction is algebraic: $\phi_T = a_T b_T / (a_T + b_T) \sim \alpha/T^2$, hence $T\phi_T \rightarrow 0$, so the expected clean switch-on count vanishes even though the Markov chain is well defined for every finite T .

7 Why feasible

The Lean route is mostly finite linear algebra plus Markov path enumeration. The repository already has `Cells.thetaHat_apply_eq_weighted_sum`, the Q1 history basis, the saturated-clean FWL bridge, explicit $T = 3$ Markov cell-weight tables, and `Mathlib/MarkovErgodicity.lean` for two-state spectral identities. Stage 0 should first formalize the deterministic Theorem 1 using finite-dimensional row spaces, kernels, and interval extrema over a box. The hard conjectural work is then reduced to bounding the event-window Gram singular values by $\Lambda_T(K)$; for $K = \{-1, 0, 1\}$, this should be checkable by explicit path enumeration analogous to the existing $T = 3$ certificates before attempting a general K induction.

8 Honest scope flags

Theorem 1 is routine linear algebra but is not yet formalized in this exact row-space/box form. Conjectures 1 and 2 are not proved. The proposal intentionally avoids claiming that exact finite- T population point identification fails when $0 < a_T, b_T < 1$; the partial and nonidentified labels are regular-identification labels after applying the deterministic threshold κ_T . The constants c_K, C_K and the exact quotient basis in Conjecture 2 may require restricting K_T growth relative to the mixing time; this is the main open technical boundary.

9 Iteration trail

- v1 (pivot) — initial draft for angle 1, replacing the dead generic spectral-rank kernel with a Sun–Abraham event-window rank-collapse kernel; prior verdict: none.

Bibliography

References

- Susan Athey and Guido W. Imbens. Design-based analysis in difference-in-differences settings with staggered adoption. *Journal of Econometrics*, 226(1):62–79, 2022.
- Iavor Bojinov, Ashesh Rambachan, and Neil Shephard. Panel experiments and dynamic causal effects: A finite population perspective. *Quantitative Economics*, 12(4):1171–1196, 2021.
- Iavor Bojinov, David Simchi-Levi, and Jinglong Zhao. Design and analysis of switchback experiments. *Management Science*, 69(7):3759–3777, 2023.
- Kirill Borusyak, Xavier Jaravel, and Jann Spiess. Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, 91(6):3253–3285, 2024.
- Brantly Callaway and Pedro H. C. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.
- Clement de Chaisemartin and Xavier D’Haultfoeuille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996, 2020.
- Clement de Chaisemartin and Xavier D’Haultfoeuille. Difference-in-differences estimators of intertemporal treatment effects. *Review of Economics and Statistics*, 2024.
- Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021.
- Sukjin Han. Identification in nonparametric models for dynamic treatment effects. *Journal of Econometrics*, 225(2):132–147, 2021.

- Yuchen Hu and Stefan Wager. Switchback experiments under geometric mixing. arXiv:2209.00197, 2022.
- Charles F. Manski. *Partial Identification of Probability Distributions*. Springer, 2003.
- Ashesh Rambachan and Jonathan Roth. A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591, 2023.
- Liyang Sun and Sarah Abraham. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199, 2021.
- Elie Tamer. Partial identification in econometrics. *Annual Review of Economics*, 2:167–195, 2010.